# Alireza Shafaei

Alireza-Shafaei | ali.shafaei.80@sharif.edu | shafaei.erfan.20019@gmail.com | (+98) 991 263 0612

## Education

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| Sep. 2024 – March 2027 (Expected) | **M.S. in Electrical Engineering (Digital Systems)**, Sharif University of Technology, Tehran, Iran.[1]<br>GPA: 18.02/20 (3.87/4) |
| Sep. 2019 – Sep. 2024 | **B.S. in Electrical Engineering**, Sharif University of Technology, Tehran, Iran.<br>GPA: 17.03/20 (3.52/4) |
| Feb. 2021 – Sep. 2024 | **Minor in Economics**, Sharif University of Technology, Tehran, Iran. |

## Research Interests

- Deep Generative Modeling
- Mathematical Optimization Theory
- Statistical Machine Learning
- Reinforcement Learning
- Large Language Models & RAG
- Edge-Cloud Computing
- MLOps
- Trustworthy & Privacy in Machine Learning
- Embedded Systems
- FPGA

## Selected Courses

- Deep Generative Models (Graduate)
- Statistical Learning (Graduate)
- Big Data Analytics (Graduate)
- Convex Optimization 2 (Graduate)
- Deep Learning (Graduate)
- Neural Networks (Graduate)
- Data communication network (Graduate)
- Embedded Systems (Graduate)
- Neuroscience of Learning, Memory, Cognition (Graduate)
- Adv. Computer Architecture (Graduate)
- Intro. to Machine Learning
- Convex Optimization 1
- Communication Circuits
- Electronics 3
- Pulse Tech. and Digital Circuits
- Design Microprocessor Systems
- Econometrics
- Financial Economics
- Digital Economics
- Macroeconomics
- Microeconomics
- System-2 AI (Graduate, Audit)
- Deep Reinforcement Learning (Graduate, Audit)

## Academic Projects & Research Experience

- **In-Context Learning as a Service: Lyapunov-Optimized Split Computing (2025–present) (master's thesis):** A quantized cloud LLM orchestrates DNN partitioning across heterogeneous edge nodes via RAG, with a deterministic non-learned guardrail certifying two-sided memory feasibility. Roofline-based GPU energy model (proved strictly convex and non-monotone in clock); NP-hard transmit-power sub-problem delegated to a fractional-programming solver with monotone convergence; Lyapunov drift-plus-penalty queue stability. Validated on a physically separated two-machine testbed with a rate-shaped uplink. Thesis in its final stage; journal manuscript in preparation for *IEEE Internet of Things Journal*. *Supervisor: Dr. Hamed Shah-Mansouri.*
- **Membership Inference in Decomposed Diffusion Models (2026–present):** First controlled privacy study of wavelet diffusion and diffusion–GAN models: 11 pretrained generators across pixel, wavelet and latent families, 13 published attacks plus CBT, our own member-free calibrated band × time test. Pre-registered design (Zenodo) with blinded splits and low-FPR evaluation. *Supervisor: Dr. Sajjad Amini.*

- **Retrain, Alert, or Rollback: A Three-Response Decision Framework for Production ML (2026):** A five-state machine giving each drift response an explicit trigger and cost, with joint retraining gates, two-stage canary deployment and cause-typed rollback. Built the evaluation harness, data-provenance pipeline and archival artifact; pre-registered protocol over 13 streams with rank-based analysis. Preprint on arXiv (2611.02251); under peer review at Data Mining and Knowledge Discovery. *Supervisors: Dr. Babak Hossein Khalaj and Dr. Hamid Hadibeik.*
- **Sense, Predict, React: A Brain-Inspired, Risk-Controlled Edge–Cloud Uplink for IoT Event Detection (2026, proposal stage):** Follow-up to the master's thesis: an IoT node with a retina-like event encoder (delta spikes, homeostatic thresholds), a spiking predictor whose error is both the uplink message and the anomaly evidence (predictive coding), and a verified local reflex. Online risk control bounds missed events under censored feedback via randomised audits, and a Lyapunov virtual queue enforces a byte budget. Matched, ablated comparison of spiking, dense and activity-sparse models on TSB-AD-M and ESA-ADB over a real-link testbed. *Supervisor: Dr. Hamed Shah-Mansouri.*
- **Reinforcement Learning and Intelligent State Discretization (2024) (bachelor's thesis):** Replaced uniform state-space discretisation in tabular Q-learning with a usage-driven refinement that recovers each task's equilibria and critical transition regions from the agent's own visitation statistics, benchmarked against PPO on CartPole, MountainCar and Pendulum. *Supervisor: Dr. Saeed Bagheri Shouraki.*

## Publications & Research Artifacts

- **Retrain, Alert, or Rollback? A Three-Response Decision Framework for Production ML Under Concept Drift**
  A. Sadeghian†, **A. Shafaei**†, H. Hadibeik, B.H. Khalaj
  *arXiv preprint* `arXiv:2611.02251` *[cs.LG], November 2026.*
  *Under peer review at Data Mining and Knowledge Discovery (Springer).* †*Equal contribution.*

- **In-Context Learning as a Service: Lyapunov-Optimized Split Computing for Edge AI**
  **A. Shafaei**, H. Shah-Mansouri
  *Manuscript in preparation; target venue IEEE Internet of Things Journal.*

- **Where Do Decomposed Diffusion Models Leak Membership?**
  **A. Shafaei**, S. Amini
  *Manuscript in preparation; target venue PoPETs 2027. Design pre-registered and time-stamped on Zenodo, October 2026:* `10.5281/zenodo.23454091`

## Skills

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| Programming | Python: (PyTorch, PySpark, Tensorflow, scikit-learn, Django, Keras, Pandas, NumPy, Matplotlib, Seaborn, SciPy, Gym, MLflow), C, Verilog (FPGA development), MATLAB |
| Research Practice | Pre-registration, blinded evaluation, bootstrap confidence intervals, equivalence testing, reproducible pipelines, artifact checksumming |
| Design & Sim | PSpice, HSPICE, Proteus, ADS, Omnet++ |
| Other | MIPS/AVR/ARM, Linux, Embedded Systems, LaTeX, Microsoft Office, Problem Solving |

## Course Projects

- **Generative Models & Deep Learning (2025–2026):**
  - Implemented and trained an Energy-Based Model (EBM) on the MNIST dataset using Langevin dynamics.
  - Developed a time-aware post-training quantization framework for diffusion language models to achieve significant model compression and inference acceleration.
  - Implemented WaveNet (autoregressive generative model for raw audio), Probabilistic Graph Forecasting, CVAE for digit generation, and normalizing flows (RealNVP, PermutFlow).
  - Built a continuous retraining loop using **Prefect** for orchestration and **MLflow** for lifecycle tracking

and model registry. Developed a **FastAPI** deployment server featuring dynamic, zero-downtime model hot-swapping. Implemented a **SciPy (KS-Test)** monitoring engine to detect data/concept drift and trigger automated retraining.

- **Advanced Machine Learning & Data Analysis (2024–2025):**
  - *Massive Data Analysis:* Designed graph/collaborative recommenders (Pixie-style, Topic-Specific PageRank, HITS) and built a PySpark Structured Streaming pipeline for real-time sentiment aggregation.
  - *Statistical Learning:* Built end-to-end pipelines for age/gender estimation (Linear/Ridge/Lasso, PCA, ROC-AUC) with a custom drag-and-drop Python GUI for bounding box face detection.
  - Implemented GAN-BERT for AI vs. Human text recognition, complex CNNs for TIG Welding classification, and Hopfield/SOM networks.
- **Networking & Embedded Systems (2024–2025):**
  - *Data Networks:* Implemented a UDP-based DNS server and verified queries in Wireshark; built socket-level SMTP clients and a mailbox-capable SMTP server with auth; implemented Data Link, MAC and Network layers in Omnet++.
  - *IoT Real-time Event Detection (Orange Pi):* Developed an edge device with LAMP dashboard, Python audio detectors, and MQTT pub/sub to a Mosquitto broker, hardened with systemd limits.
- **Hardware & Circuit Design (2019–2023):** Designed and simulated MIPS multi-cycle architecture (Verilog), high-frequency MOSFET Amplifiers and Opamp 741 (HSPICE), digital stopwatches, and multivibrators (Proteus/PSpice).
- **Early ML & Computational Modeling (2023–2024):** Hodgkin–Huxley & LIF neuron simulations, VAE/CVAE implementations, clustering with EM Algorithm for GMMs, and classical classification (SVM, Decision Trees, AlexNet Fine-tuning).

## Teaching Assistantship

- Deep Generative Models (Graduate), Dr. Amini (2026)
- Neural Networks (Graduate), Dr. Bagheri Shouraki (2025)
- Machine Learning and Vision Lab, Dr. Mohammadzadeh (2025)
- Electronics 1 & 2, Dr. Fakharzadeh (2022, 2023)
- Principles of Solid State Devices, Dr. Sarvari (2023)
- Design Microprocessor Systems, Dr. Hajsadeghi (2023)

## Work Experience

- **AI Engineer & Backend Developer**, Athlx (Startup, 2026)
  - Built scalable **Django** backends for Building Management Systems (BMS) to process real-time IoT telemetry, and developed time-series anomaly detection models for predictive maintenance and facility energy optimization.

## Honors & Awards

- Ranked **21st** among 20,000 in the Iranian M.Sc. Entrance Exam in Electrical Engineering, 2024
- Ranked **24th** among 150,000 in Iranian University Entrance Exam (Math & Physics), 2019

## Languages

Persian (native) | English (advanced) | Arabic (intermediate)

## References

**Dr. Hamed Shah-Mansouri**, Assistant Professor, Sharif Univ. of Tech., master's thesis supervisor hamedsh@sharif.edu
**Dr. Babak Hossein Khalaj**, Professor, Sharif Univ. of Tech., research supervisor khalaj@sharif.edu
**Dr. Saeed Bagheri Shouraki**, Professor, Sharif Univ. of Tech., bachelor's thesis supervisor bagheri-s@sharif.edu

[1]Ranked 151-200 in electrical engineering according to QS World University Ranking in 2026. (Ranked 1st in Iran)